

analytical_exercises/plots_em.py

```

1 import os
2 import numpy as np
3 import matplotlib.pyplot as plt
4 from mpl_toolkits.mplot3d import Axes3D # needed for 3D surface plots
5
6 # -----
7 # 0. Ensure 'plots/' directory exists
8 # -----
9 # If 'plots' does not exist, it is created. If it exists, nothing happens.
10 os.makedirs("plots", exist_ok=True)
11
12 # -----
13 # 1. Define EM parameter values
14 # -----
15 #  $\theta(0) = (\theta_1(0), \theta_2(0)) = (2, 3)$ 
16 theta0_1 = 2.0 # rate of exponential before EM
17 theta0_2 = 3.0 # upper bound of uniform before EM
18
19 #  $\theta(1) = (\theta_1(1), \theta_2(1)) = (3/7, 5)$ 
20 theta1_1 = 3.0 / 7.0 # rate of exponential after EM
21 theta1_2 = 5.0 # upper bound of uniform after EM
22
23 # -----
24 # 2. Define x1 and x2 ranges
25 # -----
26 # x1 from 0 to 6 (covers samples 1, 2, 4 and the exponential tail)
27 x1 = np.linspace(0, 6, 200)
28
29 # x2 ranges:
30 # - Before EM: [0, 3]
31 # - After EM : [0, 5]
32 x2_before = np.linspace(0, theta0_2, 150)
33 x2_after = np.linspace(0, theta1_2, 150)
34
35 # Create 2D grids (mesh) of x1 and x2 values
36 X1_before, X2_before = np.meshgrid(x1, x2_before)
37 X1_after, X2_after = np.meshgrid(x1, x2_after)
38
39 # -----
40 # 3. Define p(x1, x2) before and after EM
41 # -----
42 # Because  $p(x_1, x_2) = p(x_1) * p(x_2)$ , we compute marginals and multiply.
43
44 # --- Before EM:  $\theta(0) = (2, 3)$  ---
45 #  $p(x_1 | \theta_1(0)) = 2 * \exp(-2 x_1)$ ,  $x_1 \geq 0$ 
46 p1_before = theta0_1 * np.exp(-theta0_1 * X1_before)
47
48 #  $p(x_2 | \theta_2(0)) = 1/3$ ,  $0 \leq x_2 \leq 3$ 
49 # (we already restrict x2_before to [0, 3])
50 p2_before = (1.0 / theta0_2) * np.ones_like(X2_before)
51

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52 # Joint density (properly normalized on  $[0, \infty) \times [0, 3]$ )
53 p_before = p1_before * p2_before
54
55 # --- After EM:  $\theta(1) = (3/7, 5)$  ---
56 #  $p(x_1 \mid \theta(1)) = (3/7) * \exp(-(3/7) x_1)$ ,  $x_1 \geq 0$ 
57 p1_after = theta1_1 * np.exp(-theta1_1 * X1_after)
58
59 #  $p(x_2 \mid \theta(1)) = 1/5$ ,  $0 \leq x_2 \leq 5$ 
60 p2_after = (1.0 / theta1_2) * np.ones_like(X2_after)
61
62 # Joint density (properly normalized on  $[0, \infty) \times [0, 5]$ )
63 p_after = p1_after * p2_after
64
65 # -----
66 # 4. Plot the two surfaces side by side
67 # -----
68 fig = plt.figure(figsize=(10, 4))
69
70 # Before EM ( $\theta(0)$ )
71 ax1 = fig.add_subplot(1, 2, 1, projection='3d')
72 ax1.plot_surface(X1_before, X2_before, p_before)
73 ax1.set_title('p(x1, x2) at  $\theta(0) = (2, 3)$ ')
74 ax1.set_xlabel('x1')
75 ax1.set_ylabel('x2')
76 ax1.set_zlabel('density')
77
78 # After EM ( $\theta(1)$ )
79 ax2 = fig.add_subplot(1, 2, 2, projection='3d')
80 ax2.plot_surface(X1_after, X2_after, p_after)
81 ax2.set_title('p(x1, x2) at  $\theta(1) = (3/7, 5)$ ')
82 ax2.set_xlabel('x1')
83 ax2.set_ylabel('x2')
84 ax2.set_zlabel('density')
85
86 plt.tight_layout()
87
88 # -----
89 # 5. Save figure inside 'plots/' directory
90 # -----
91 output_path = "plots/EM_Plots.png"
92 fig.savefig(output_path)
93
```